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We will discuss several questions from Chapter 5 in Ross' book.

Tutorial Questions

- (1) Two types of coins are produced at a factory: a fair coin and a biased one that comes up heads 55 percent of the time. We have one of these coins, but do not know whether it is a fair coin or a biased one. In order to ascertain which type of coin we have, we shall perform the following statistical test: We shall toss the coin 1000 times. If the coin lands on heads 525 or more times, then we shall conclude that it is a biased coin, whereas if it lands on heads less than 525 times, then we shall conclude that it is a fair coin. If the coin is actually fair, what is the probability that we shall reach a false conclusion? What would it be if the coin were biased?
- (2) (a) A fire station is to be located along a road of length A , $A < \infty$. If fires occur at points uniformly chosen on $(0, A)$, where should the station be located so as to minimize the expected distance from the fire? That is, choose a so as to

$$\text{minimize } \mathbb{E}[|X - a|]$$
 when X is uniformly distributed over $(0, A)$.
 - (b) Now suppose that the road is of infinite length—stretching from point 0 outward to ∞ . If the distance of a fire from point 0 is exponentially distributed with rate λ , where should the fire station now be located? That is, we want to minimize $\mathbb{E}[|X - a|]$, where X is now exponential with rate λ .
- (3) Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait
 - (a) more than 10 minutes;
 - (b) between 10 and 20 minutes.
- (4) If X is an exponential random variable with parameter $\lambda = 1$, compute the probability density function of the random variable Y defined by $Y = \ln X$.
- (5) If X has a probability density f_X . Find the density function of $Y = |X|$.
- (6) Let

$$f(x) = \frac{\theta^2}{1 + \theta}(1 + x)e^{-\theta x}, \quad x > 0$$

where $\theta > 0$.

- (a) Show that f is a probability density function.
- (b) If X has density function f , find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Assignment Questions

- (1) The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 1/2$. What is
 - (a) the probability that a repair time exceeds 2 hours?
 - (b) the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours?
- (2) Consider a post office that is staffed by two clerks. Suppose that when Mr. Smith enters the system, he discovers that Ms. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Ms. Jones or Mr. Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with parameter λ , what is the probability that, of the three customers, Mr. Smith is the last to leave the post office?
- (3) (a) Let X be a continuous random variable with cumulative distribution function F . Define the random variable Y by $Y = F(X)$. Show that Y is uniform on $(0, 1)$.
 (b) Let Y be a uniform random variable on $(0, 1)$ and F be a continuous increasing function. Define the random variable X by $X = F^{-1}(Y)$, where $F^{-1}(z)$ is defined as

$$F^{-1}(z) = \min\{x : z \leq F(x)\}.$$

Show that the CDF of X is exactly F .

Remark: Part (b) is useful in generating (pseudo-)random numbers in a computer system. One can just apply F^{-1} to a uniform (pseudo-)random number, which is relatively easy to generate. This is practical as long as F^{-1} can be calculated easily.

- (c) Show how to generate an exponential random variable of parameter λ using a random variable on $(0, 1)$.
- (d) Show how to generate a random variable with the following density function:

$$f(x) = \begin{cases} e^{2x}, & \text{if } -\infty < x < 0; \\ e^{-2x}, & \text{if } 0 \leq x < \infty. \end{cases}$$

- (4) Let X be a normal random variable with mean μ and variance σ^2 . Find the density function of $Y = e^X$.

Remark: Y is called lognormal random variable with parameters μ and σ^2 . The lognormal is often used as the distribution of the ratio of the price of a security at the end of the day to its price at the end of the prior day.

- (5) Let

$$f(x) = \begin{cases} \frac{1}{3}e^x & \text{if } x < 0; \\ \frac{1}{3} & \text{if } 0 \leq x < 1; \\ \frac{1}{3}e^{-(x-1)} & \text{if } x \geq 1. \end{cases}$$

- (a) Show that f is a probability density function.
- (b) If X has density function f , find $\mathbb{E}(X)$.
- (6) If X is uniformly distributed over $(0, 1)$, find the density function of $Y = e^X$.